System Requirements

- **Operating System**: Windows, macOS, or Linux
- **Python**: Version 3.6 or higher
- **Pygame**: Version 2.0 or higher
- **Hardware**: Standard desktop or laptop with a mouse (trackpads may reduce precision)
- **Display**: Minimum resolution of 1280x720 for optimal experience

Installation

1. **Install Python**:
    - Download and install Python 3.6+ from python.org.
    - Ensure pip is included in your Python installation.
2. **Install Pygame**:
    - Open a terminal or command prompt and run: pip install pygame
3. **Download the Game Files**:
    - Obtain the five Python files: aim_trainer.py, ui_manager.py, target.py, particle.py, and button.py.
    - Place all files in the same directory.
4. **Run the Game**:
    - Navigate to the game directory in your terminal or command prompt.
    - Run the main script:

python aim_trainer.py

    - The game window (1280x720) should appear, centered on your screen.

How to Play

Starting the Game

1. **Launch**: Run aim_trainer.py to open the game.
2. **Main Menu**: You'll see the "Ultimate Aim Trainer" title with three buttons:
    - **Start**: Begins the game with the default difficulty (Medium).
    - **Options**: Opens the difficulty selection menu.
    - **Quit**: Exits the game.
3. **Options Menu**: Choose a difficulty (Easy, Medium, Hard) or select "Back" to return to the main menu. Selecting a difficulty starts the game immediately.

4. **Game Start Prompt**: After selecting "Start" or a difficulty, a "Click anywhere to begin!" prompt appears. Click anywhere in the game window to start the 60-second game session.

Game Objective

- **Primary Goal**: Click as many targets as possible within 60 seconds or before losing all 10 lives to achieve the highest score and accuracy.
- **Secondary Goal**: Build combos by hitting targets quickly to increase your score multiplier, maximizing your points.

Controls

- **Mouse Left-Click**: Click on targets to score hits. Also used to interact with menu buttons.
- **ESC Key**: Pauses the game, opening a menu with "Resume" and "Quit" options.
- **Mouse Movement**: Move the cursor (a white crosshair with a black outline) to aim at targets.

Gameplay Mechanics

- **Target Spawning**: Circular targets (red and white concentric circles) spawn at random positions on the screen, avoiding the edges (30-pixel padding) and the top bar (50 pixels high). Targets pulse slightly (5% size oscillation) for visual feedback.
- **Target Behavior**: Each target grows to a maximum size (25 pixels radius) and then shrinks, disappearing if not clicked. The speed of growth/shrink depends on the difficulty and increases over time.
- **Hitting Targets**: Click a target to score a hit, which increments your score and spawns up to 20 particle effects (yellow/orange circles that fade out over 1 second).
- **Missing Targets**: Clicking outside a target counts as a miss, reducing your lives by 1 and triggering a brief screen shake (10 frames, ±5 pixels).
- **Game Duration**: The game lasts 60 seconds or ends early if you lose all 10 lives.
- **Pause Menu**: Press ESC to pause, showing "Resume" (continue playing) or "Quit" (exit to main menu). Transitions between menus and gameplay use smooth fade effects.

Difficulty Levels

The game offers three difficulty levels, adjustable via the Options menu:

- **Easy**:
    - Targets spawn every 500ms.
    - Growth/shrink speed: 0.2 (slower changes).
    - Ideal for beginners or casual practice.
- **Medium** (default):
    - Targets spawn every 400ms.

    - Growth/shrink speed: 0.3 (moderate challenge).
    - Suitable for intermediate players.
- **Hard**:
    - Targets spawn every 300ms.
    - Growth/shrink speed: 0.4 (faster changes).
    - Designed for advanced players seeking a challenge.
- **Dynamic Scaling**: As the game progresses, targets grow and shrink faster, calculated as base_speed * (1 + elapsed_time / 60), making the game increasingly difficult.

Scoring and Combos

- **Scoring**: Each target hit awards points equal to your current multiplier (e.g., x3 multiplier gives 3 points per hit).
- **Combo System**: Hitting targets within 0.5 seconds of each other increases your combo count, which sets your multiplier (e.g., 3 consecutive hits yield a x3 multiplier).
- **Miss Penalty**: Missing a target resets the combo and multiplier to 1, requiring you to rebuild your streak.
- **Example**: If you hit 3 targets in quick succession, your multiplier becomes x3, and the third hit awards 3 points. A miss resets the multiplier to x1.

Lives System

- **Lives**: You start with 10 lives, displayed as "Lives: X" in the top bar.
- **Misses**: Each miss (clicking outside a target) reduces lives by 1 and triggers a screen shake for feedback.
- **Game Over**: Losing all 10 lives ends the game immediately, displaying "Game Over!" on the end screen.

User Interface

- **Main Menu**: Features an animated title (bouncing slightly) and buttons with hover effects (10% scale increase).
- **Top Bar**: During gameplay, a semi-transparent bar at the top displays real-time stats:
    - Time: Format MM:SS.m (e.g., 01:23.4).
    - Speed: Hits per second (e.g., 1.2 t/s).
    - Hits: Total targets hit.
    - Lives: Remaining lives (10 - misses).
    - Multiplier: Current score multiplier (e.g., x3).

    - Combo: Current streak of consecutive hits.
    - Score: Total points earned.
- **Cursor**: A white crosshair with a black outline follows your mouse, ensuring precise aiming.
- **Background**: A gradient background transitions from dark blue (0, 25, 40) at the top to a lighter shade (10, 20, 40) at the bottom, enhancing visual appeal.
- **Particles**: Hits spawn colorful particles (yellow, orange) that move outward and fade, adding satisfying feedback.

Game Over and End Screen

- **End Conditions**:
    - **Time's Up!**: After 60 seconds, the game ends, showing your final stats.
    - **Game Over!**: Losing all 10 lives ends the game early.
- **End Screen**: Displays:
    - Time: Total playtime (e.g., 00:60.0).
    - Hits: Number of targets hit.
    - Clicks: Total mouse clicks.
    - Accuracy: (hits / clicks) * 100, rounded to one decimal (e.g., 82.7%).
    - Score: Total points, including multiplier effects.
- **Options**: A "Quit" button returns to the main menu. (Note: A "Restart" option is not currently available but may be added in future updates.)
- **Purpose**: The end screen provides a clear summary of your performance, helping you track improvement.

Tips for Success

1. **Aim for Combos**: Click targets quickly (within 0.5 seconds) to build your combo and multiplier, significantly boosting your score.
2. **Time Your Clicks**: Click when targets are largest (around 25 pixels radius) for easier hits, as they shrink quickly, especially on Hard difficulty.
3. **Stay Calm on Misses**: A miss resets your multiplier, but don't rush—focus on precision to avoid further lives loss.
4. **Choose the Right Difficulty**: Start with Easy to learn the mechanics, then progress to Medium or Hard as your skills improve.
5. **Use a Mouse**: A trackpad can work but may reduce accuracy. A gaming mouse with adjustable DPI is ideal for precision.

6. **Practice Regularly**: Short, focused sessions (e.g., 5–10 games) can improve your reaction speed and accuracy over time.
7. **Watch the Top Bar**: Monitor your lives and multiplier to adjust your strategy (e.g., play safer if lives are low).

Troubleshooting

- **Game Doesn't Start**:
    - Ensure Python and Pygame are installed correctly (pip show pygame).
    - Verify all five .py files are in the same directory.
    - Check for Python errors in the terminal and ensure Python 3.6+ is used.
- **Lag or Low FPS**:
    - Close background applications to free up system resources.
    - Ensure your system meets the minimum requirements.
    - Reduce particle effects by editing aim_trainer.py (e.g., lower the particle cap from 100).
- **Mouse Clicks Not Registering**:
    - Check your mouse for hardware issues.
    - Ensure no other applications are capturing mouse input.
    - Try a different USB port or mouse.
- **Window Not Centered**:
    - Verify os.environ['SDL_VIDEO_CENTERED'] = '1' is in aim_trainer.py.
    - Check your display settings for resolution compatibility.